

2 D Heat Equation (steady-state)

Monday, July 14, 2025 4:24 AM

7_ Steady-State 2D Heat Equation

7.1 Two dimensional heat-equation

$$\frac{\partial u}{\partial t} = c^2 \nabla^2 u$$

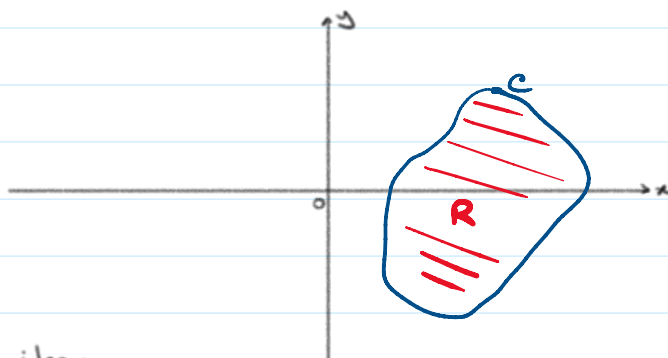
Here, we will look at steady-state solutions where $\frac{\partial u}{\partial t} = 0$.

So the Heat Equation becomes: $c^2 \nabla^2 u = 0$

i.e. $\boxed{\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0}$ Laplace's Equation

7.2 Dirichlet Problem:

Suppose R is a region in the xy -plane with boundary curve C



General idea:

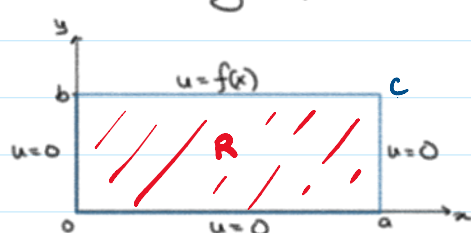
If C is parametrized as $\vec{r}(s) = [x(s), y(s)]$, $a \leq s \leq b$

and we want to solve: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

with BC: $u(x(s), y(s)) = f(s)$

Then this is a Dirichlet problem.

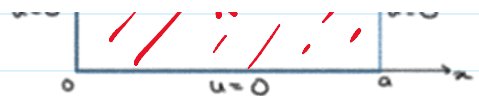
7.3 We will solve a very simple case: Dirichlet Problem in a Rectangle R .



Solve: $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$

with BC: $u(x, b) = f(x)$

$u(x, 0) = u(0, y) = u(a, y) = 0$ } for (x, y) on the boundary C of R



$$u(x,0)=u(0,y)=u(a,y)=0 \quad \text{boundary } C \text{ of } R$$

7.4 Use Separation of Variables to obtain two ODEs from $\nabla^2 u = 0$

Let $u(x,y) = F(x)G(y)$, then $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$ becomes

$$\frac{\partial^2}{\partial x^2} F(x)G(y) + \frac{\partial^2}{\partial y^2} F(x)G(y) = 0$$

$$\Rightarrow G(y) \frac{d^2 F}{dx^2} + F(x) \frac{d^2 G}{dy^2} = 0$$

$$\Rightarrow \frac{1}{F(x)} F''(x) = -\frac{1}{G(y)} G''(y) = -k$$

Then $\frac{1}{F(x)} F''(x) = -k \Rightarrow F''(x) + kF(x) = 0$

and $-\frac{1}{G(y)} G''(y) = -k \Rightarrow G''(y) = kG(y) \Rightarrow G''(y) - kG(y) = 0$

7.5 For the ODE $F''(x) + kF(x) = 0$, what are the boundary conditions?
What solution do we get for $F(x)$?

Since $u=0$ at $x=0$ and $x=a$, the boundary conditions are:

$$F(0)=0 \quad \text{and} \quad F(a)=0$$

As discussed for the wave equation, this means $k = \omega^2 > 0$.

so the DE is: $F''(x) + \omega^2 F(x) = 0$

with solutions $F(x) = A \cos(\omega x) + B \sin(\omega x)$

Just as with the wave equation: $F(0)=0 \Rightarrow A=0$

and $F(a)=0 \Rightarrow B \sin(\omega a) = 0 \Rightarrow \omega = \frac{n\pi}{a}$

Thus:

$$F(x) = B \sin\left(\frac{n\pi}{a} x\right)$$

7.6 What are the boundary conditions for $G(y)$, and what solution do we get?

Since $u=0$ at $y=0$ and $u=f(x)$ at $y=b$, then

Since $u=0$ at $y=0$ and $u=f(x)$ at $y=b$, then

$G(0)=0$, and $G(b)$ is undetermined at this point

Now from: $\frac{d^2 G}{dy^2} - kG(y)=0$. we get

$$G''(y) - \left(\frac{n\pi}{a}\right)^2 G(y) = 0 \quad \left[\text{Characteristic equation: } m^2 - \left(\frac{n\pi}{a}\right)^2 = 0 \right. \\ \left. \Rightarrow m = \pm \left(\frac{n\pi}{a}\right) \right]$$

which has solutions of the form:

$$G(y) = P e^{\frac{n\pi}{a}y} + Q e^{-\frac{n\pi}{a}y}$$

From $G(0)=0$, get

$$P + Q = 0 \Rightarrow P = -Q$$

So then

$$G(y) = P \left[e^{\frac{n\pi}{a}y} - e^{-\frac{n\pi}{a}y} \right]$$

$$\Rightarrow G(y) = 2P \sinh\left(\frac{n\pi y}{a}\right)$$

7.7 What would we get if we put $F(x)$ and $G(y)$ in $u(x,y)=F(x)G(y)$?

$$\text{We get } u(x,y) = B \sin\left(\frac{n\pi x}{a}\right) \cdot 2P \sinh\left(\frac{n\pi y}{a}\right)$$

$$\text{i.e. } u(x,y) = B_n \sin\left(\frac{n\pi x}{a}\right) \sinh\left(\frac{n\pi y}{a}\right)$$

7.8 How do we resolve the final BC: $u(x,b)=f(x)$ for $0 \leq x \leq a$?

$$\text{We consider } u_n(x,y) = B_n \sin\left(\frac{n\pi x}{a}\right) \sinh\left(\frac{n\pi y}{a}\right)$$

$$\text{and let } u(x,y) = \sum_{n=1}^{\infty} u_n(x,y)$$

Then $u(x,b)=f(x)$ becomes

$$\sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi x}{a}\right) \sinh\left(\frac{n\pi b}{a}\right) = f(x) \quad [\text{Fourier Sine Series}]$$

$$\text{In that case: } B_n \sinh\left(\frac{n\pi b}{a}\right) = \frac{2}{a} \int_0^a f(x) \sin\left(\frac{n\pi x}{a}\right) dx$$

$$\text{i.e. } B_n = \frac{2}{a \sinh\left(\frac{n\pi b}{a}\right)} \int_0^a f(x) \sin\left(\frac{n\pi x}{a}\right) dx$$

7.9 Thus, the Dirichlet problem:

$$\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} = 0$$

$$\left. \begin{array}{l} \text{with BC: } u(0, \frac{y}{b}) = u(a, \frac{y}{b}) = u(\frac{x}{a}, 0) = 0 \\ \text{and} \\ u(\frac{x}{a}, b) = f(\frac{x}{a}) \end{array} \right\} \begin{array}{l} \text{Here the boundary curve } C \\ \text{has been parameterized piecewise} \\ \text{for } 0 \leq s \leq 1. \end{array}$$

has solution:

$$u(x, y) = \sum_{n=1}^{\infty} B_n \sin\left(\frac{n\pi}{a}x\right) \sinh\left(\frac{n\pi}{a}y\right)$$

$$\text{with } B_n = \frac{2}{a \sinh(n\pi b/a)} \int_a^a f(x) \sin\left(\frac{n\pi}{a}x\right) dx$$